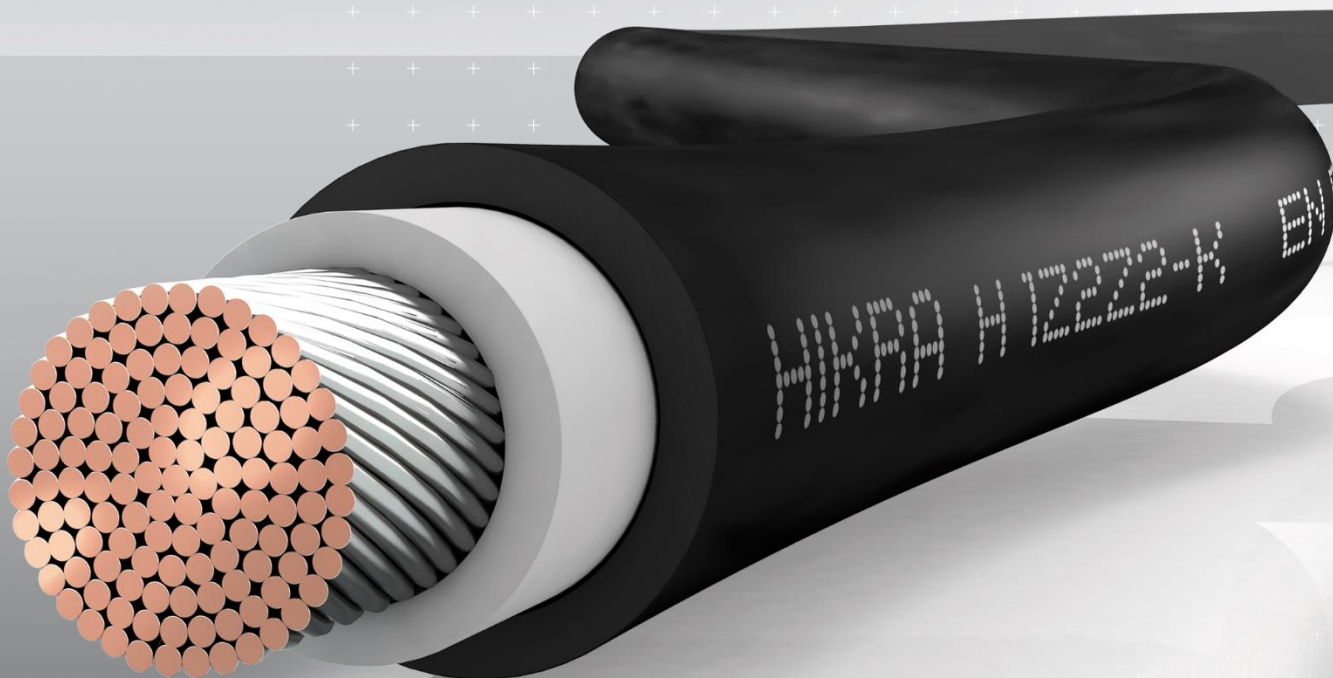


HIKRA®

solar cables
part of HIS CONNECT™

HIKRA® SOL EN50618 (H1Z2Z2-K) IEC62930; CPR: B2_{ca} s1a,d0,a1

DATA SHEET



HIKRA® Solar Cable



This cable is intended for use in PV installations e.g. acc. IEC 60364-7-712 and suitable for the application in/at equipment with protective insulation (protection class II). Our double insulated HIKRA® SOL 1500V solar cable is intended for permanent use outdoor and indoor, for free movable, free hanging and fixed installation. Robust materials defy the long-term influences of nature and offer a maximum quality and safety. In case of proper use, the expected lifetime of this product will be at least 25 years. Thanks to its good resistance to atmospheric conditions, the cable has a specifically water repellency and can be laid underground. Please refer to our laying instructions.

IN FOCUS IS THE PLANT REVENUE. IN OPERATION OUR SOLAR CABLES.

HIKRA® stands for technically outstanding and integrated solutions, which have been developed taking into account all environmental, safety and costs aspect and even under the harshest conditions delivering a faultless performance.

Certification

- EN50618; H1Z2Z2-K; 50363076
- IEC62930; R 5048873
- RoHS & Reach conform



YOUR ADVANTAGES

- Electron-beam cross-linked special compound: XLPO/XLPO
- Higher water resistance
- Direct burial
- CPR-classification of burning behavior: B2ca
- Nominal voltage: Max. 1,8kV DC operating voltage
- Control consumption: We ship this cable with meter-marking
- Compatibility to all major PV connectors
- Individual packaging units: 100m coils, 500 m drums or in another kind upon request
- Global availability. Comprehensive international logistic concept

HIKRA®: The comprehensive cabling solution for international successful EPC and Installation companies, which expect more from their suppliers.

TECHNICAL DATA

Construction	
Strand construction	Tin-plated copper strand (electrolytic copper), fine wire acc. IEC 60228 Class 5
Insulation	Electron-beam cross-linked Polyolefin; Shore hardness D 32
Outer Sheath	Electron-beam cross-linked special compound XLPO; Shore hardness D 36
Colour	Sheath: black, red; Insulation: clear – naturally colored
Marking	HIKRA SOL1500V H12222-K IEC62930 1x6,0mm ² R 50363076 CE with meter marking

Technical characteristics	
Nominal voltage	1,5kV DC and 1,0kV AC
Maximum permitted operating voltage:	1,8kV DC (2,0 kV DC internal examination)
Voltage test on complete cable	6,5kV AC / 15kV DC (5 minutes water bath, 20±5°C)
Current carrying capacity	See document „Current rating – HIKRA® Solar Cable“ November 2013
Max. resistance of conductor	EN 50395 Clause 5
Short-circuit-temperature	250° C/5s

Material properties	
UV stability	Tensile strength and ultimate-elongation after 720 h (360 cycles) ≥ 70% of initial values; EN 50289-4-17 acc. Method A; EN ISO 4892-1 (2000) and EN ISO 4892-2 (2006)
Ozone resistance	72h, relative humidity 55±5%, Temperature 40±2°C (EN 50396 Method B; Ozone concentration (200±50)x10 ⁻⁶)
Insulation resistance	Insulation resistance in water bath, each 2h at +90°C and 2h at 20°C (Limit values acc. EN 50618 Table 1)
Surface resistance of sheath	≥10 ⁹ Ω (applied voltage 100-500V DC, 1 minute) acc. EN 50395 Clause 11
Dynamic penetration test	Spring-steel-needle through insulation or sheath (EN50618 Annex D)
Direct burial	Water immersion at 90°C, duration 12 weeks; Insulation resistance ≥ 3GΩ (internal examination acc. UL44 cl. 5.4 & UL2556 6.4.4.2.1
Crushing- and impact-resistance	Impact-Resistance UL 854.23 and Crushing-Resistance UL 854.24 (internal examination)
Sheath resistance against acid and alkaline	168h at 23°C in N-Oxal acid and N-Sodium hydroxide (EN 60811-404); ammoniac-resistant
Behavior in case of fire	Flame-retardant acc. EN 60332-1-2 Annex A, low smoke emission (EN 61034,-2)
CPR-Performance	Fire behavior acc. to No. 305/2011/EU: B2ca s1a,d0,a1 acc. EN 50575:2014
Halogen-free	EN 50525-1, Annex B
Cold impact test	EN 60811-506, EN 50618 Annex C.1 at -40°C
Cold bending test	-40°C;16h (EN 60811-504)
Cold elongation test	Max. 30% elongation at -40±2°C, 16h (EN 60811-505)
Damp heat test	Duration 1000h at 90°C and min. 85% relative humidity (EN 60068-2-78)
Minimum bending radius flexible / fixed	10x cable diameter 4x cable diameter

Temperature Range	
Temperature	Ambient temperature: -40° C to +90°C; Maximum conductor temperature: +120° C
Maximum storage temperature	+40°C
Minimum temperature for installation and handling	-25°C

Order No. black	Order No. red	Cross-section mm ²	Construction n x max.-Ø (mm)	Max. resistance (Ω/km)	External diameter (+/- 0,2 mm)	Copper index kg/km	Approx. Weight kg/km
739065	739066	1 x 1.5	29 x 0.25	13.7	4.6	14.0	32.0
738609	738610	1 x 2.5	47 x 0.25	8.21	5.0	24.0	42.0
738613	738614	1 x 4.0	52 x 0.3	5.09	5.4	38.4	57.0
738615	738616	1 x 6.0	78 x 0.3	3.39	6.0	57.6	76.0
738617	738618	1 x 10.0	77 x 0.4	1.95	7.2	96.0	119.0
738619	-	1 x 16.0	126 x 0.4	1.24	9.3	153.6	196.0
739061	-	1 x 25.0	190 x 0.4	0.795	11.3	240.0	291.0

HISCONNECT®

— solar wiring system



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HIS Renewables GmbH
We reserve the right to change product information
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